S. Nitheesh
192524329
Cloud Computing
CSA 1501

CO4-AT1

① Unified Big Data Architecture:

Use data ingestion → distributed storage → processing → Analytics → recommendation. Tools like Kafka, Hadoop / HDFS, Spark, & NoSQL can integrate online and offline customer data. Real time analytics helps provide personalized recommendations.

② Distributed Computing:

Distributed Systems divide large datasets among multiple machine. Parallel Processing allows tasks to run simultaneously, reducing processing time and improving scalability.

③ Fraud Detection:

Use Kafka for real time, transaction, Spark streaming for processing, & ML models for fraud prediction. Streaming detects suspicious transactions immediately, with predictive models identify unusual patterns.

(4) Multimedia Big Data:

Video, Audio, and image requires large storage high bandwidth, & heavy processing. Use cloud/object storage, HDFS, NoSQL, & distributed processing such as Spark for efficient management.

(5) Disaster Recovery:

Use Data replication, redundant servers, backups and automated failover. If one system fails, another replica takes over, ensuring business continuity & high availability.

(6) Data Quality:

Inconsistency, duplication, and missing values produce inaccurate analytics. Use data validation, cleaning, deduplication, metadata management and data governance to improve reliability.

(7) Recommendation System:

Collect user behaviour such as watch history, rating, & searches, Distributed technologies like spark process large datasets, while ML algorithms generates personalized recommendations & improve engagement.

(8) Cloud computing in Big Data:

Cloud provides scalability, flexibility pay-as-you-use pricing, and easy resource manangement. Limitations include security/privacy risks, dependency on internet connectivity, and possible cost increases

(9) Transportation Analytics:

Collect GPS, traffic, vehicles, and fuel data using IoT and streaming system. Big Data analytics finds optimal routes, reduces fuel consumption, and supports real-time operational decisions.

(10) Ethical & Legal challenges:

Major issues include Privacy, data security consent, bias, and unauthorized used. Organizations should implement access control, encryption, transparency, consent policies, audits, and regulatory compliance to maintain users trust.